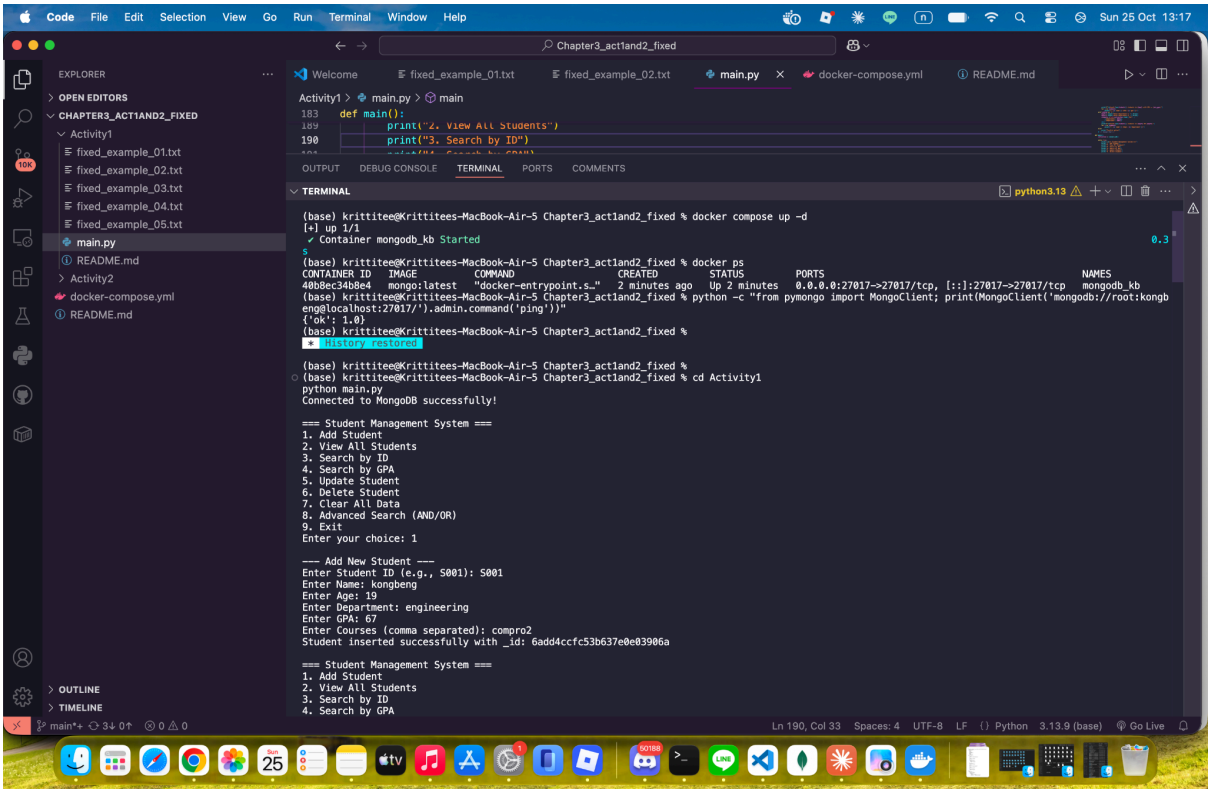
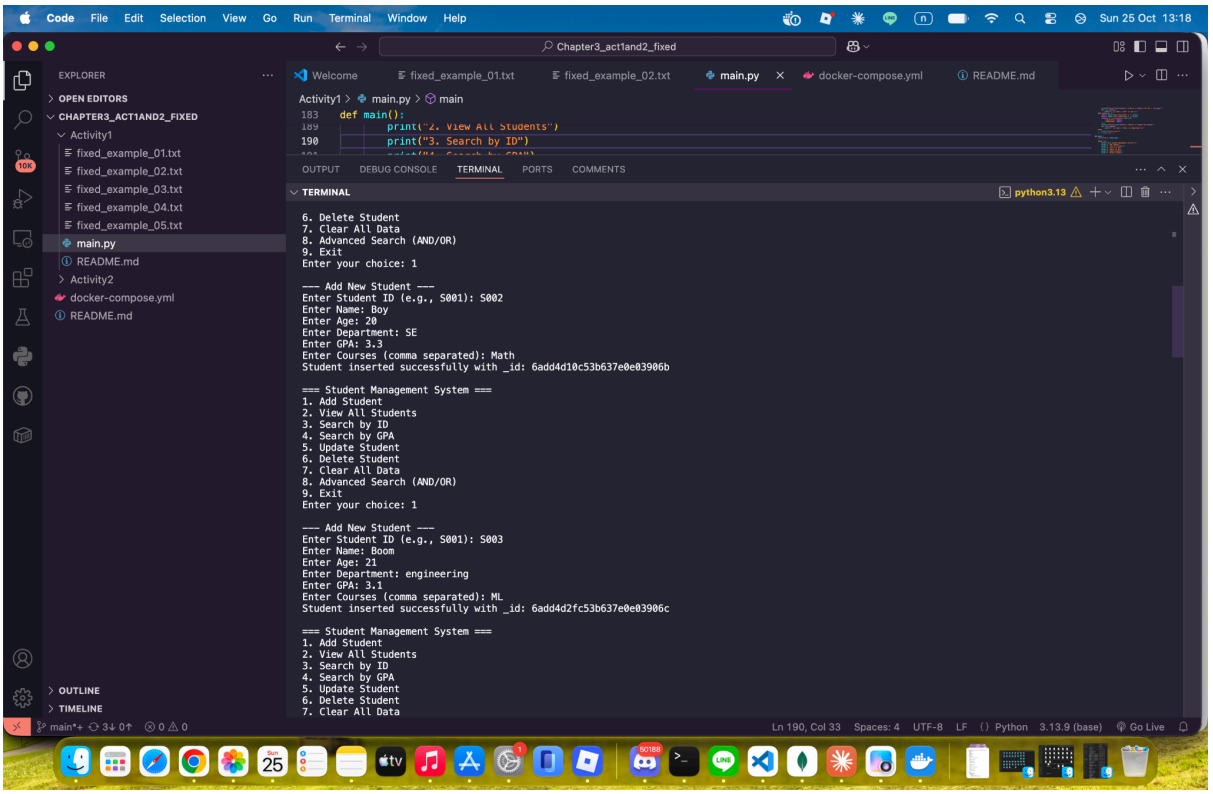
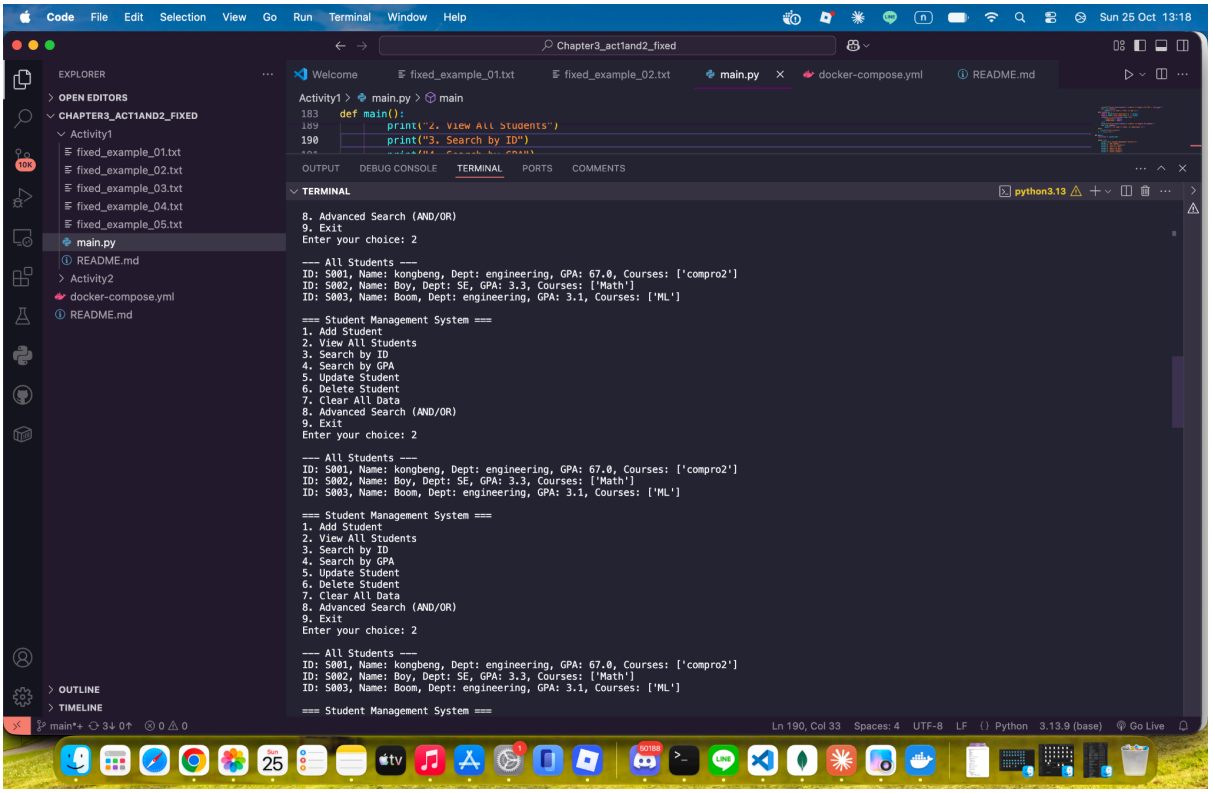
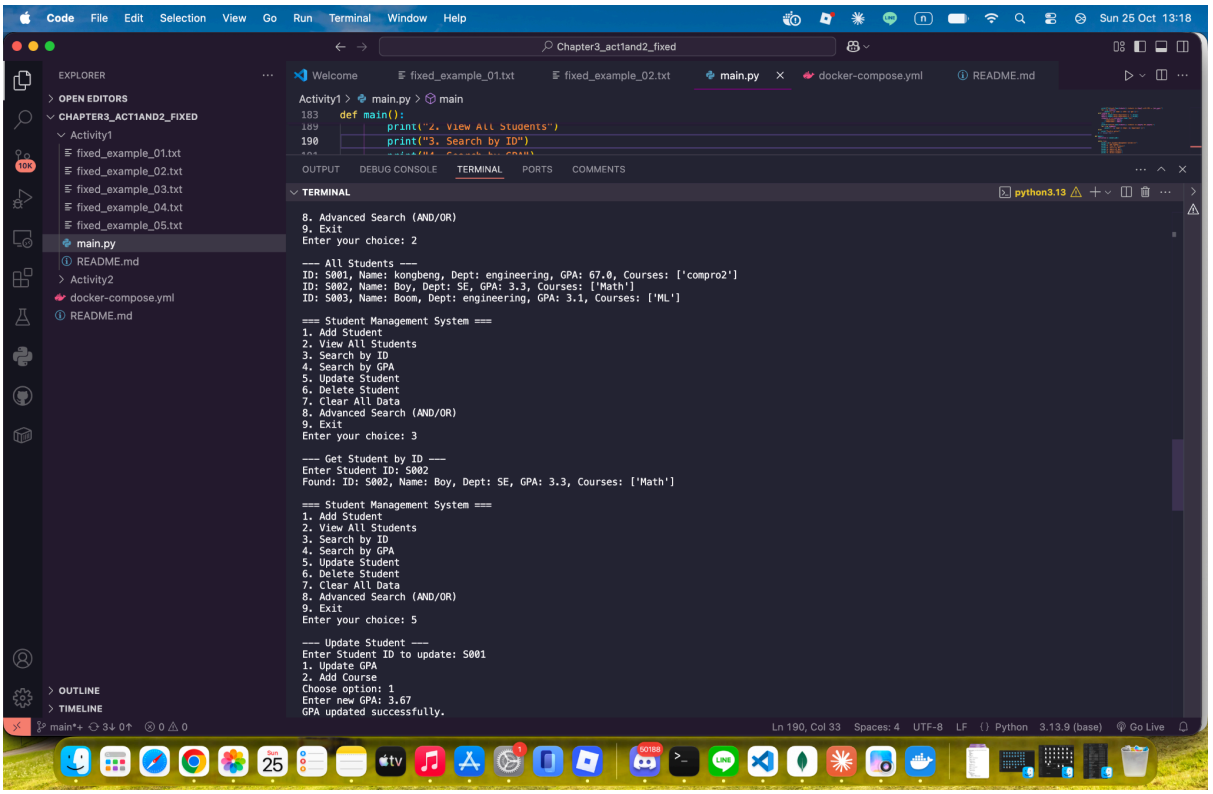
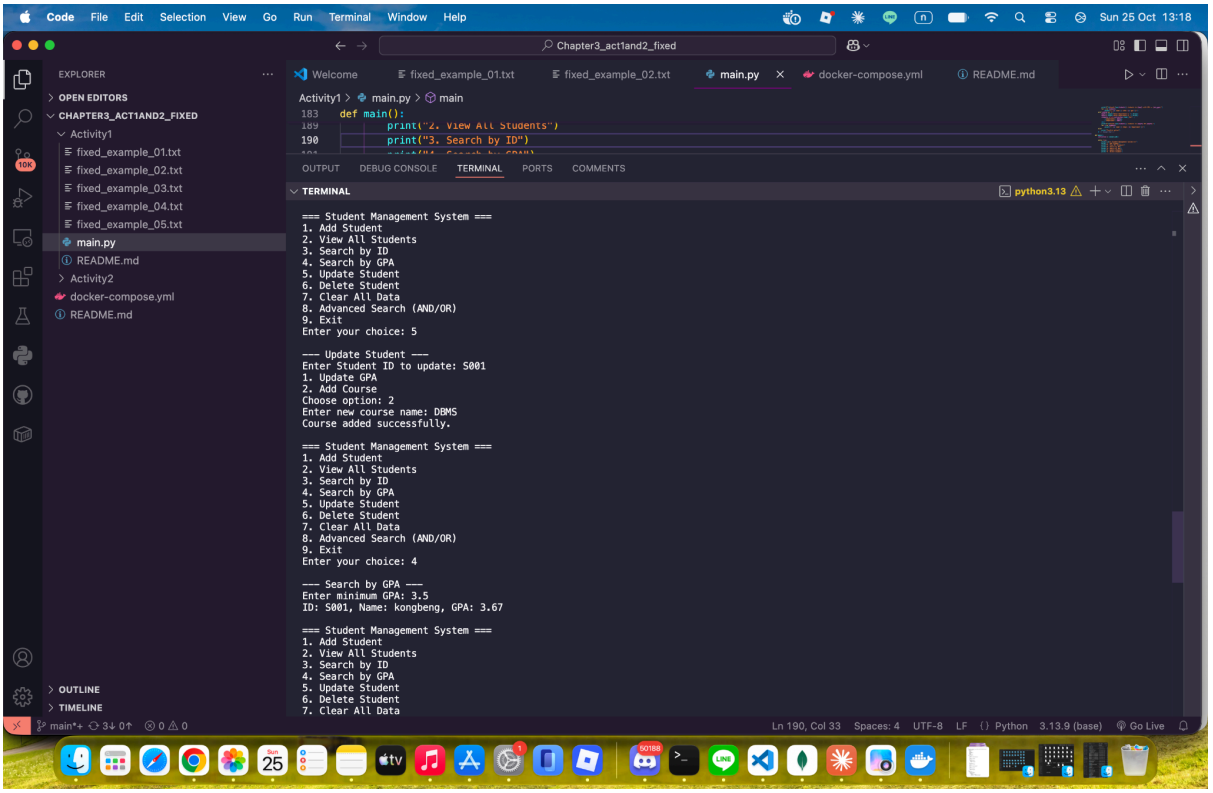
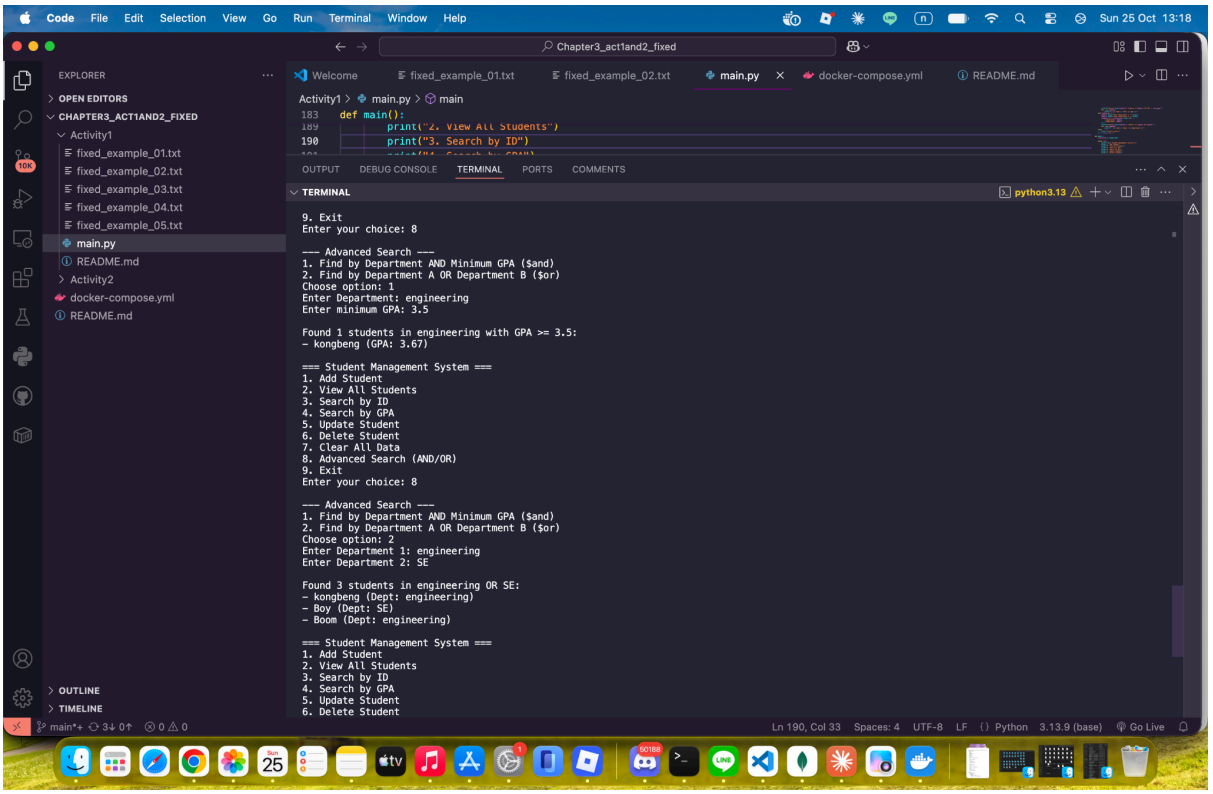

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 13:17
Chapter3_act1and2_fixed
EXPLORER
OPEN EDITORS
CHAPTER3_ACT1AND2_FIXED
Activity1
fixed_example_01.txt
fixed_example_02.txt
fixed_example_03.txt
fixed_example_04.txt
fixed_example_05.txt
main.py
README.md
Activity2
docker-compose.yml
README.md
Welcome
fixed_example_01.txt
fixed_example_02.txt
main.py
docker-compose.yml
README.md
Activity1 > main.py > main
183 def main():
190 print("3. Search by ID")
OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
TERMINAL
python3.13
(base) krittiteeg@Krittitees-MacBook-Air-5 Chapter3_act1and2_fixed % docker compose up -d
[+] up 1/1
✓ Container mongodb_kb Started
0.3
5
(base) krittiteeg@Krittitees-MacBook-Air-5 Chapter3_act1and2_fixed % docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
40b8ec34b8e4 mongo:latest "docker-entrypoint.s…" 2 minutes ago Up 2 minutes 0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp mongodb_kb
(base) krittiteeg@Krittitees-MacBook-Air-5 Chapter3_act1and2_fixed % python -c "from pymongo import MongoClient; print(MongoClient('mongodb://root:kongbeng@localhost:27017/').admin.command('ping'))"
{'ok': 1.0}
(base) krittiteeg@Krittitees-MacBook-Air-5 Chapter3_act1and2_fixed %
History restored
(base) krittiteeg@Krittitees-MacBook-Air-5 Chapter3_act1and2_fixed %
(base) krittiteeg@Krittitees-MacBook-Air-5 Chapter3_act1and2_fixed % cd Activity1
python main.py
Connected to MongoDB successfully!
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 1
--- Add New Student ---
Enter Student ID (e.g., S001): S001
Enter Name: kongbeng
Enter Age: 19
Enter Department: engineering
Enter GPA: 67
Enter Courses (comma separated): compro2
Student inserted successfully with _id: 6add4ccfc53b637e0e03906a
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
OUTLINE
TIMELINE
main*+ 3↓ 0↑ 0 0
Ln 190, Col 33 Spaces: 4 UTF-8 LF Python 3.13.9 (base) Go Live

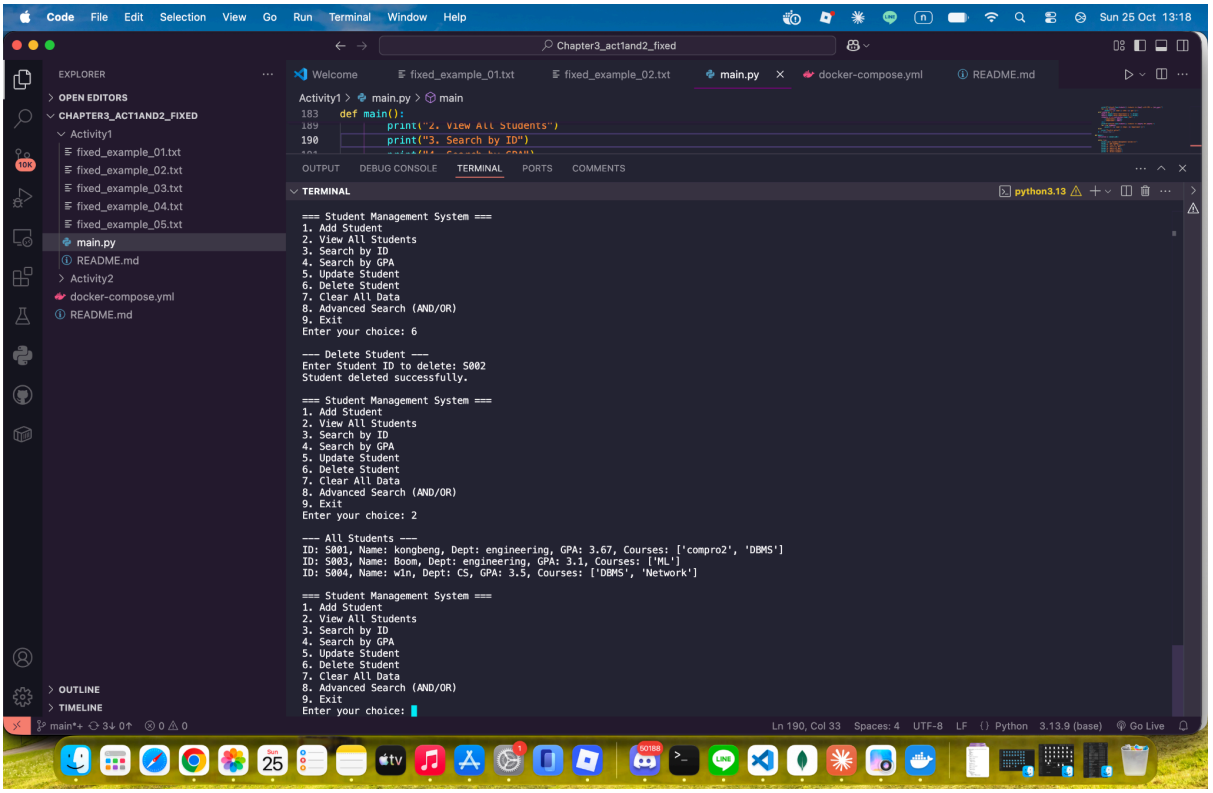

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 13:18
Chapter3_act1and2_fixed
EXPLORER
OPEN EDITORS
CHAPTER3_ACT1AND2_FIXED
Activity1
fixed_example_01.txt
fixed_example_02.txt
fixed_example_03.txt
fixed_example_04.txt
fixed_example_05.txt
main.py
README.md
Activity2
docker-compose.yml
README.md
Welcome
fixed_example_01.txt
fixed_example_02.txt
main.py
docker-compose.yml
README.md
Activity1 > main.py > main
183 def main():
190 print("3. Search by ID")
OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
TERMINAL
python3.13
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 1
--- Add New Student ---
Enter Student ID (e.g., S001): S002
Enter Name: Boy
Enter Age: 20
Enter Department: SE
Enter GPA: 3.3
Enter Courses (comma separated): Math
Student inserted successfully with _id: 6add4d10c53b637e0e03906b
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 1
--- Add New Student ---
Enter Student ID (e.g., S001): S003
Enter Name: Boom
Enter Age: 21
Enter Department: engineering
Enter GPA: 3.1
Enter Courses (comma separated): ML
Student inserted successfully with _id: 6add4d2fc53b637e0e03906c
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
OUTLINE
TIMELINE
main*+ 3↓ 0↑ 0 0
Ln 190, Col 33 Spaces: 4 UTF-8 LF Python 3.13.9 (base) Go Live

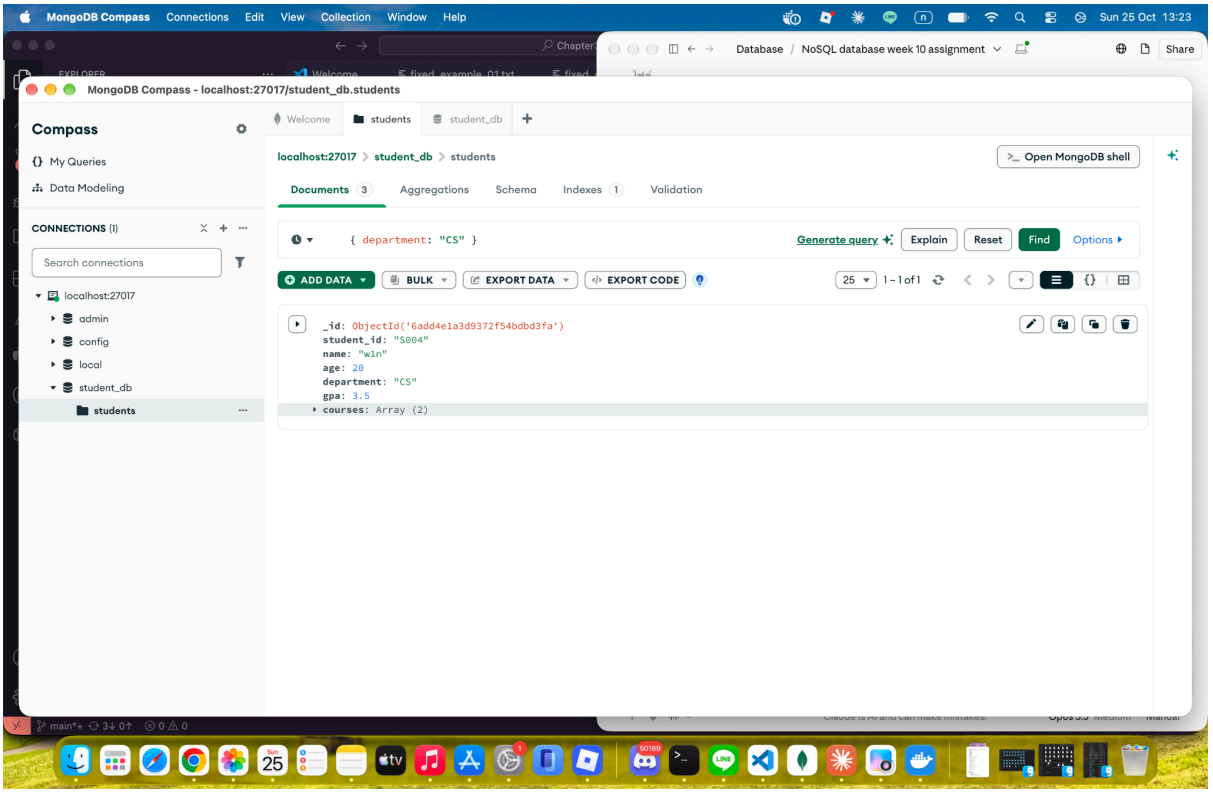

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 13:18
Chapter3_act1and2_fixed
EXPLORER
OPEN EDITORS
CHAPTER3_ACT1AND2_FIXED
Activity1
fixed_example_01.txt
fixed_example_02.txt
fixed_example_03.txt
fixed_example_04.txt
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main.py
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docker-compose.yml
README.md
Welcome
fixed_example_01.txt
fixed_example_02.txt
main.py
docker-compose.yml
README.md
Activity1 > main.py > main
183 def main():
190 print("3. Search by ID")
OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
TERMINAL
python3.13
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 2
--- All Students ---
ID: S001, Name: kongbeng, Dept: engineering, GPA: 67.0, Courses: ['compro2']
ID: S002, Name: Boy, Dept: SE, GPA: 3.3, Courses: ['Math']
ID: S003, Name: Boom, Dept: engineering, GPA: 3.1, Courses: ['ML']
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 2
--- All Students ---
ID: S001, Name: kongbeng, Dept: engineering, GPA: 67.0, Courses: ['compro2']
ID: S002, Name: Boy, Dept: SE, GPA: 3.3, Courses: ['Math']
ID: S003, Name: Boom, Dept: engineering, GPA: 3.1, Courses: ['ML']
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 2
--- All Students ---
ID: S001, Name: kongbeng, Dept: engineering, GPA: 67.0, Courses: ['compro2']
ID: S002, Name: Boy, Dept: SE, GPA: 3.3, Courses: ['Math']
ID: S003, Name: Boom, Dept: engineering, GPA: 3.1, Courses: ['ML']
=== Student Management System ===
OUTLINE
TIMELINE
main*+ 34 0 0 0
Ln 190, Col 33 Spaces: 4 UTF-8 LF Python 3.13.9 (base) Go Live

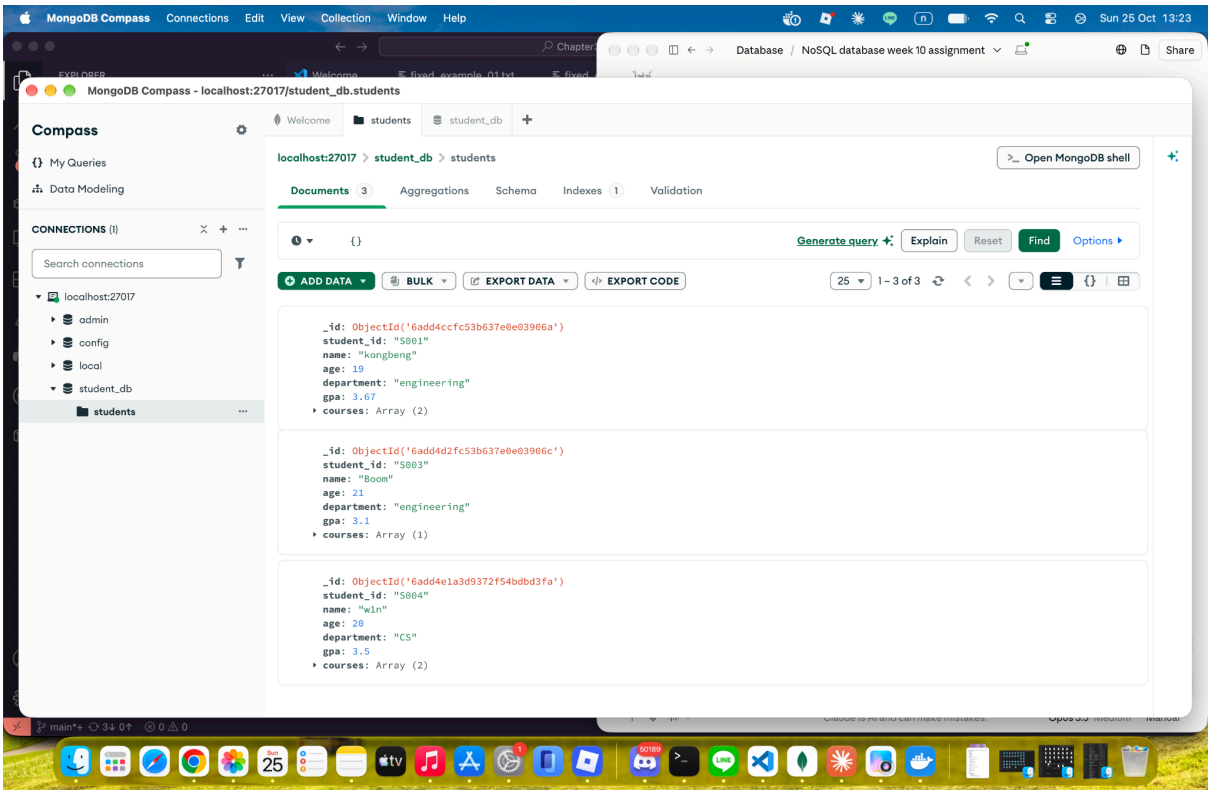

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 13:18
Chapter3_act1and2_fixed
EXPLORER
OPEN EDITORS
CHAPTER3_ACT1AND2_FIXED
Activity1
fixed_example_01.txt
fixed_example_02.txt
fixed_example_03.txt
fixed_example_04.txt
fixed_example_05.txt
main.py
README.md
Activity2
docker-compose.yml
README.md
Welcome
fixed_example_01.txt
fixed_example_02.txt
main.py
docker-compose.yml
README.md
Activity1 > main.py > main
183 def main():
190 print("3. Search by ID")
OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
TERMINAL
python3.13
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 2
--- All Students ---
ID: S001, Name: kongbeng, Dept: engineering, GPA: 67.0, Courses: ['compro2']
ID: S002, Name: Boy, Dept: SE, GPA: 3.3, Courses: ['Math']
ID: S003, Name: Boom, Dept: engineering, GPA: 3.1, Courses: ['ML']
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 3
--- Get Student by ID ---
Enter Student ID: S002
Found: ID: S002, Name: Boy, Dept: SE, GPA: 3.3, Courses: ['Math']
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 5
--- Update Student ---
Enter Student ID to update: S001
1. Update GPA
2. Add Course
Choose option: 1
Enter new GPA: 3.67
GPA updated successfully.
OUTLINE
TIMELINE
main*+ 34 0 0 0
Ln 190, Col 33 Spaces: 4 UTF-8 LF Python 3.13.9 (base) Go Live

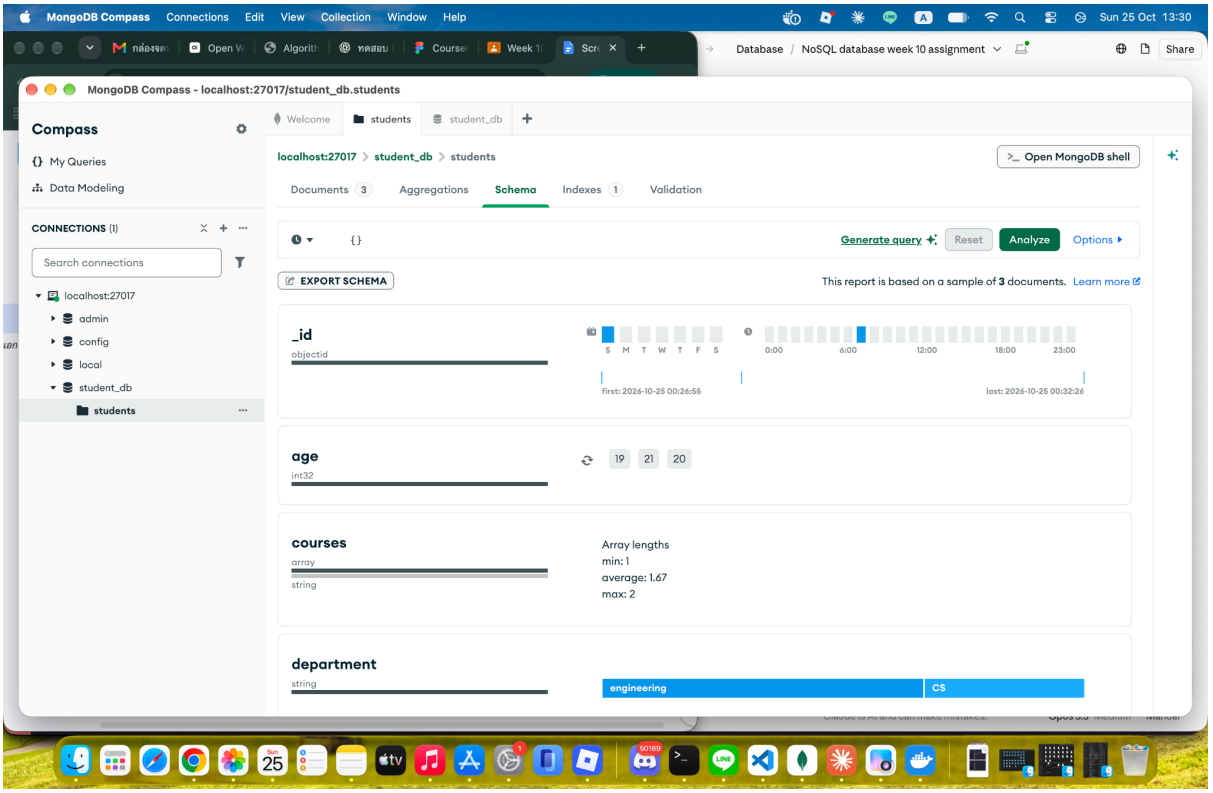

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 13:18
Chapter3_act1and2_fixed
EXPLORER
OPEN EDITORS
CHAPTER3_ACT1AND2_FIXED
Activity1
fixed_example_01.txt
fixed_example_02.txt
fixed_example_03.txt
fixed_example_04.txt
fixed_example_05.txt
main.py
README.md
Activity2
docker-compose.yml
README.md
Welcome
fixed_example_01.txt
fixed_example_02.txt
main.py
docker-compose.yml
README.md
Activity1 > main.py > main
183 def main():
189 print("2. View All Students")
190 print("3. Search by ID")
OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
TERMINAL
python3.13
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 5
--- Update Student ---
Enter Student ID to update: S001
1. Update GPA
2. Add Course
Choose option: 2
Enter new course name: DBMS
Course added successfully.
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 4
--- Search by GPA ---
Enter minimum GPA: 3.5
ID: S001, Name: kongbeng, GPA: 3.67
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
OUTLINE
TIMELINE
main*+ 3↓ 0↑ 0 0
Ln 190, Col 33 Spaces: 4 UTF-8 LF Python 3.13.9 (base) Go Live

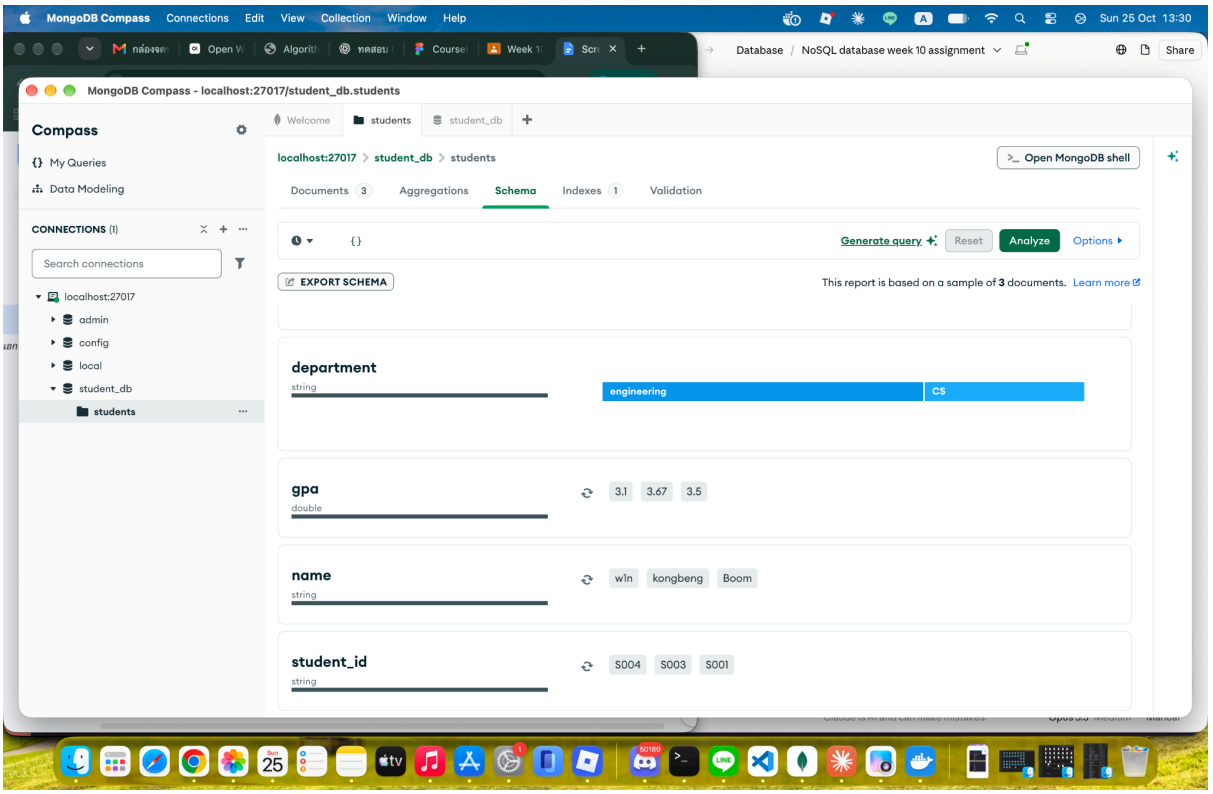

Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 13:18
Chapter3_act1and2_fixed
EXPLORER
OPEN EDITORS
CHAPTER3_ACT1AND2_FIXED
Activity1
fixed_example_01.txt
fixed_example_02.txt
fixed_example_03.txt
fixed_example_04.txt
fixed_example_05.txt
main.py
README.md
Activity2
docker-compose.yml
README.md
Welcome
fixed_example_01.txt
fixed_example_02.txt
main.py
docker-compose.yml
README.md
Activity1 > main.py > main
183 def main():
189 print("2. View All Students")
190 print("3. Search by ID")
OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
TERMINAL
python3.13
9. Exit
Enter your choice: 8
--- Advanced Search ---
1. Find by Department AND Minimum GPA ($and)
2. Find by Department A OR Department B ($or)
Choose option: 1
Enter Department: engineering
Enter minimum GPA: 3.5
Found 1 students in engineering with GPA >= 3.5:
- kongbeng (GPA: 3.67)
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 8
--- Advanced Search ---
1. Find by Department AND Minimum GPA ($and)
2. Find by Department A OR Department B ($or)
Choose option: 2
Enter Department 1: engineering
Enter Department 2: SE
Found 3 students in engineering OR SE:
- kongbeng (Dept: engineering)
- Boy (Dept: SE)
- Boom (Dept: engineering)
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
OUTLINE
TIMELINE
main*+ 3↓ 0↑ 0 0
Ln 190, Col 33 Spaces: 4 UTF-8 LF Python 3.13.9 (base) Go Live


Code File Edit Selection View Go Run Terminal Window Help
Sun 25 Oct 13:18
Chapter3_act1and2_fixed
EXPLORER
OPEN EDITORS
CHAPTER3_ACT1AND2_FIXED
Activity1
fixed_example_01.txt
fixed_example_02.txt
fixed_example_03.txt
fixed_example_04.txt
fixed_example_05.txt
main.py
README.md
Activity2
docker-compose.yml
README.md
Welcome
fixed_example_01.txt
fixed_example_02.txt
main.py
docker-compose.yml
README.md
Activity1 > main.py > main
def main():
print("2. View All Students")
print("3. Search by ID")
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
COMMENTS
TERMINAL
python3.13
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 6
--- Delete Student ---
Enter Student ID to delete: S002
Student deleted successfully.
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice: 2
--- All Students ---
ID: S001, Name: kongbeng, Dept: engineering, GPA: 3.67, Courses: ['compro2', 'DBMS']
ID: S003, Name: Boom, Dept: engineering, GPA: 3.1, Courses: ['ML']
ID: S004, Name: win, Dept: CS, GPA: 3.5, Courses: ['DBMS', 'Network']
=== Student Management System ===
1. Add Student
2. View All Students
3. Search by ID
4. Search by GPA
5. Update Student
6. Delete Student
7. Clear All Data
8. Advanced Search (AND/OR)
9. Exit
Enter your choice:
OUTLINE
TIMELINE
main*+
Ln 190, Col 33
Spaces: 4
UTF-8
LF
Python
3.13.9 (base)
Go Live


MongoDB Compass - localhost:27017/student_db.students
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
config
local
student_db
students
Welcome
students
student_db
localhost:27017 > student_db > students
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
{ department: "CS" }
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 1 of 1
_id: ObjectId('6add4e1a3d9372f54bdbd3fa')
student_id: "S004"
name: "Win"
age: 20
department: "CS"
gpa: 3.5
courses: Array (2)


MongoDB Compass - localhost:27017/student_db.students
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
config
local
student_db
students
Welcome
students
student_db
localhost:27017 > student_db > students
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
{}
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 3 of 3
_id: ObjectId('6add4ccfc53b637e0e03906a')
student_id: "S001"
name: "kongbeng"
age: 19
department: "engineering"
gpa: 3.67
courses: Array (2)
_id: ObjectId('6add4d2fc53b637e0e03906c')
student_id: "S003"
name: "Boom"
age: 21
department: "engineering"
gpa: 3.1
courses: Array (1)
_id: ObjectId('6add4e1a3d9372f54bdbd3fa')
student_id: "S004"
name: "Win"
age: 20
department: "CS"
gpa: 3.5
courses: Array (2)


MongoDB Compass - localhost:27017/student_db.students
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
config
local
student_db
students
Welcome
students
student_db
localhost:27017 > student_db > students
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
Generate query
Reset
Analyze
Options
EXPORT SCHEMA
This report is based on a sample of 3 documents. Learn more
_id
objectid
age
int32
19
21
20
courses
array
string
Array lengths
min: 1
average: 1.67
max: 2
department
string
engineering
CS


MongoDB Compass - localhost:27017/student_db.students
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
config
local
student_db
students
Welcome
students
student_db
localhost:27017 > student_db > students
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
Generate query
Reset
Analyze
Options
EXPORT SCHEMA
This report is based on a sample of 3 documents. Learn more
department
string
engineering
CS
gpa
double
3.1
3.67
3.5
name
string
wln
kongbeng
Boom
student_id
string
S004
S003
S001

Task 7: Schema Export & Analysis

1) JSON Structure (from exported schema, 3 documents)
- _id : ObjectId (auto-generated, unique)
- student_id : String (e.g., "S001")
- name : String
- age : Int32
- department : String
- gpa : Double
- courses : Array of String (length 1-2, avg 1.67)
All 7 fields appear in every document (listed as "required").

2) Equivalent RDBMS Structure
"courses" is a multi-valued attribute, so it cannot be stored in one column (violates 1NF). It must be split into a separate table.

```
CREATE TABLE students (
    student_id  VARCHAR(10)  PRIMARY KEY,
    name        VARCHAR(100) NOT NULL,
    age         INT,
    department  VARCHAR(50),
    gpa         DECIMAL(3,2)
);

CREATE TABLE student_courses (
    student_id  VARCHAR(10),
    course_name VARCHAR(50),
    PRIMARY KEY (student_id, course_name),
    FOREIGN KEY (student_id) REFERENCES students(student_id)
        ON DELETE CASCADE
);
```

3) Comparison
- MongoDB embeds courses inside each student document, so one query returns everything. RDBMS needs a JOIN between 2 tables.
- MongoDB adds _id (ObjectId) automatically; in SQL, student_id is used as the primary key instead.
- MongoDB is schema-on-read: a new field can be added to one document without ALTER TABLE. SQL is schema-on-write: every row must follow the table definition.
- Adding a course: MongoDB uses $push on the array; SQL uses INSERT INTO student_courses.

Task 6: Compass vs Python syntax
- Compass filter : { department: "CS" }
- Python (PyMongo): collection.find({"department": "CS"})
Both use the same query document; Python requires quoted keys because it is a dict.